COMPUTER ENGINEERING

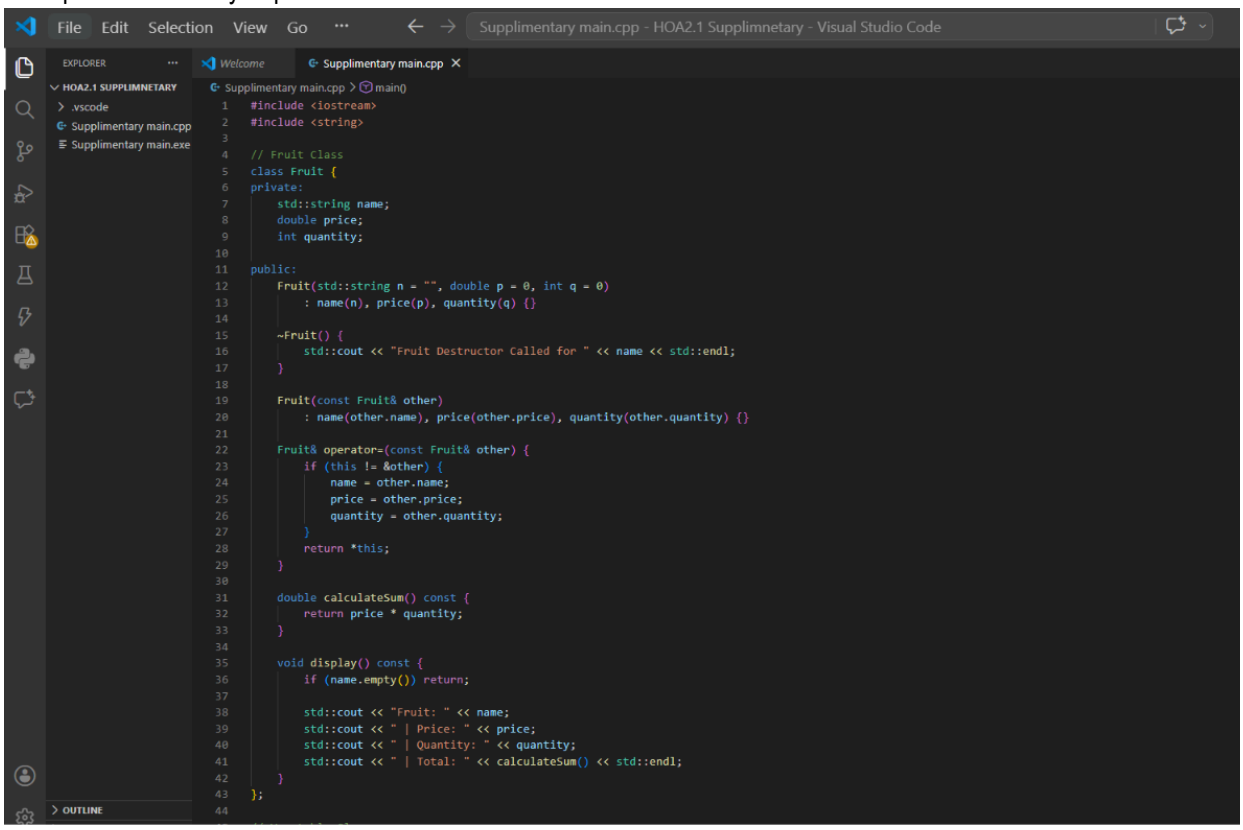
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures and Algorithm	**SECTION:**	CPE21S2
DATE PERFORMED:	07/17/2006	**DATE SUBMITTED:**	09/17/20206
NAME:	Nieles, Cenndy M.	**INSTRUCTOR:**	Engr. Jimlordlord Quejao

ACTIVITY NO. 2.1 ARRAYS, POINTER AND DYNAMIC MEMORY ALLOCATION SUPPLEMENTARY

1. PROCEDURE OUTPUT

What procedures did you perform?




```cpp
#include <iostream>
#include <string>

// Fruit Class
class Fruit {
private:
    std::string name;
    double price;
    int quantity;

public:
    Fruit(std::string n = "", double p = 0, int q = 0)
        : name(n), price(p), quantity(q) {}

    ~Fruit() {
        std::cout << "Fruit Destructor Called for " << name << std::endl;
    }

    Fruit(const Fruit& other)
        : name(other.name), price(other.price), quantity(other.quantity) {}

    Fruit& operator=(const Fruit& other) {
        if (this != &other) {
            name = other.name;
            price = other.price;
            quantity = other.quantity;
        }
        return *this;
    }

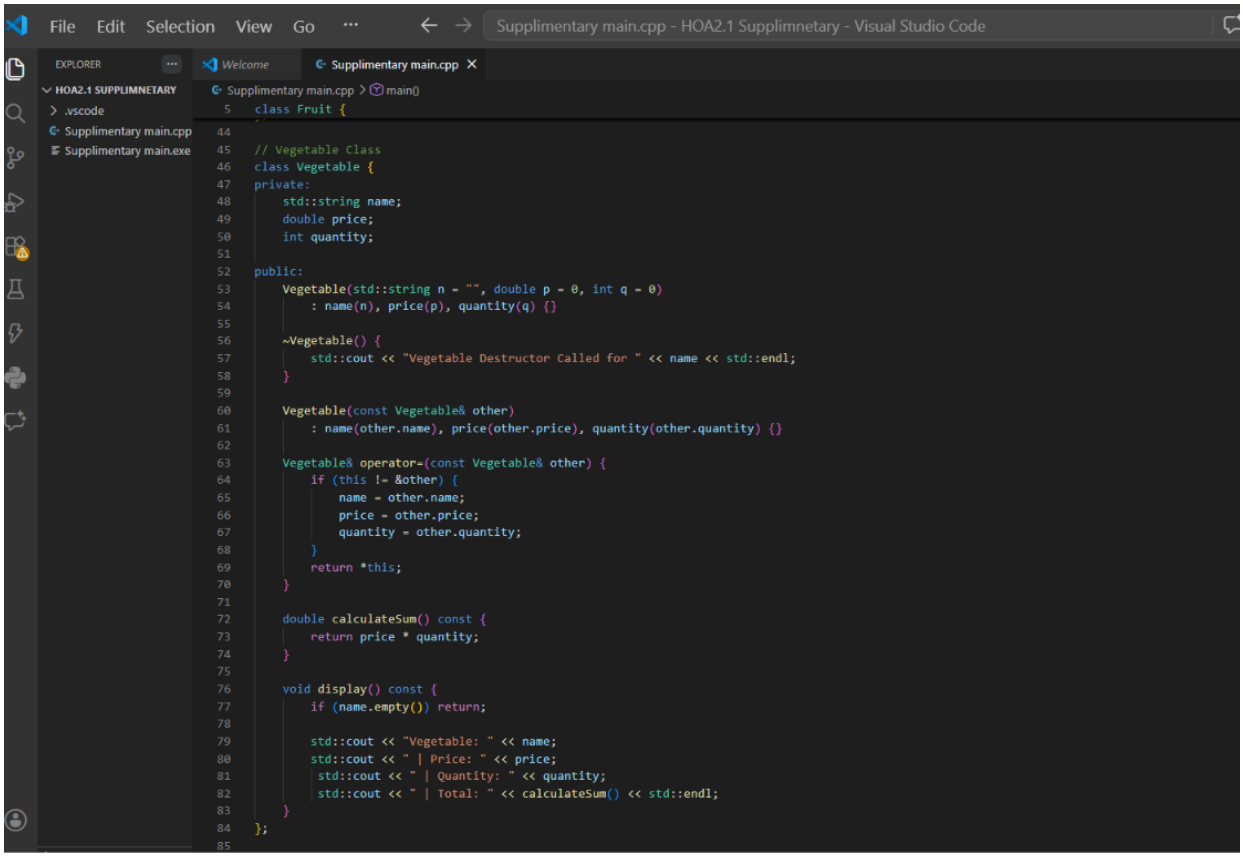
    double calculateSum() const {
        return price * quantity;
    }

    void display() const {
        if (name.empty()) return;

        std::cout << "Fruit: " << name;
        std::cout << " | Price: " << price;
        std::cout << " | Quantity: " << quantity;
        std::cout << " | Total: " << calculateSum() << std::endl;
    }
};
```

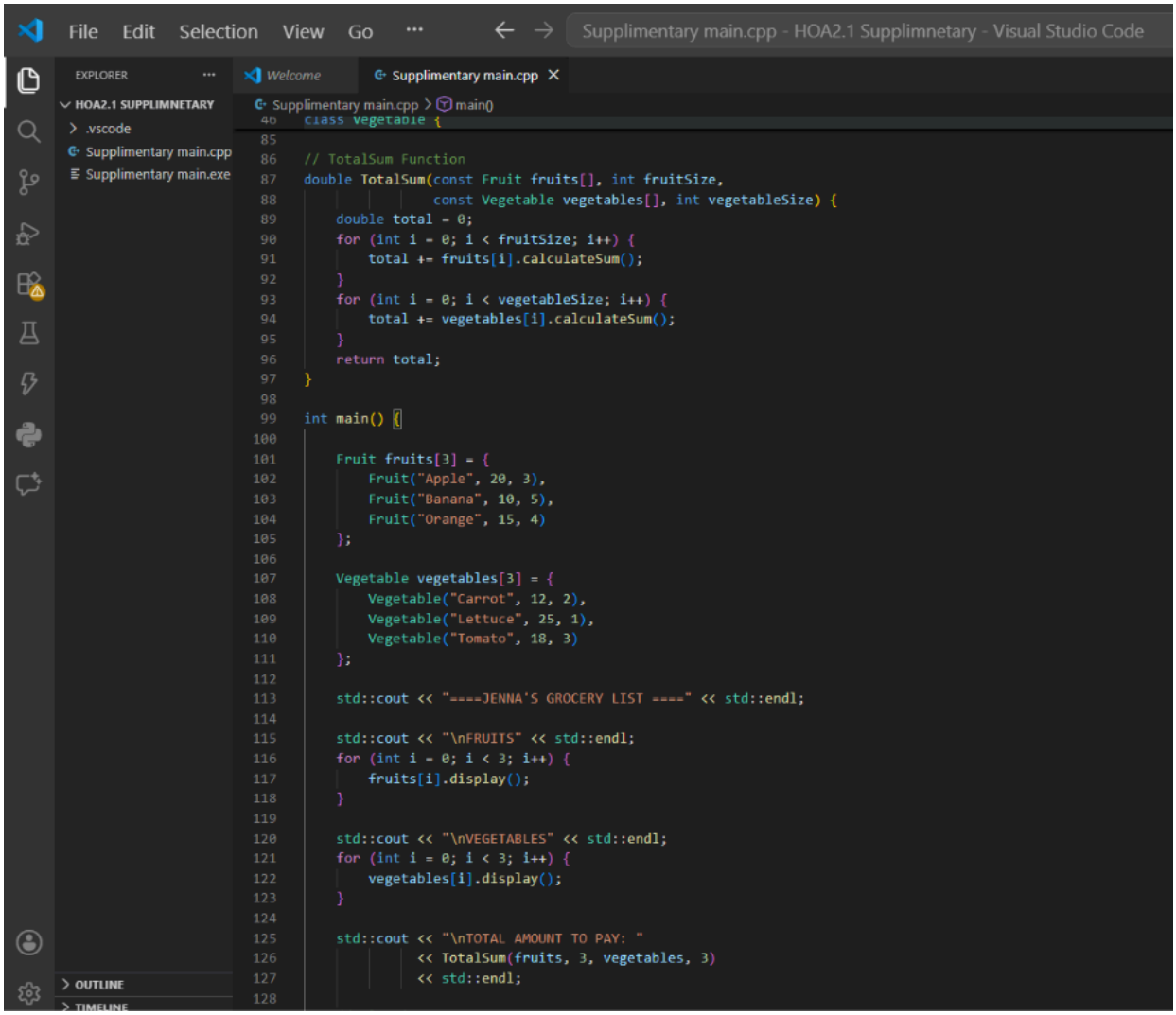
COMPUTER ENGINEERING

QUEZON CITY



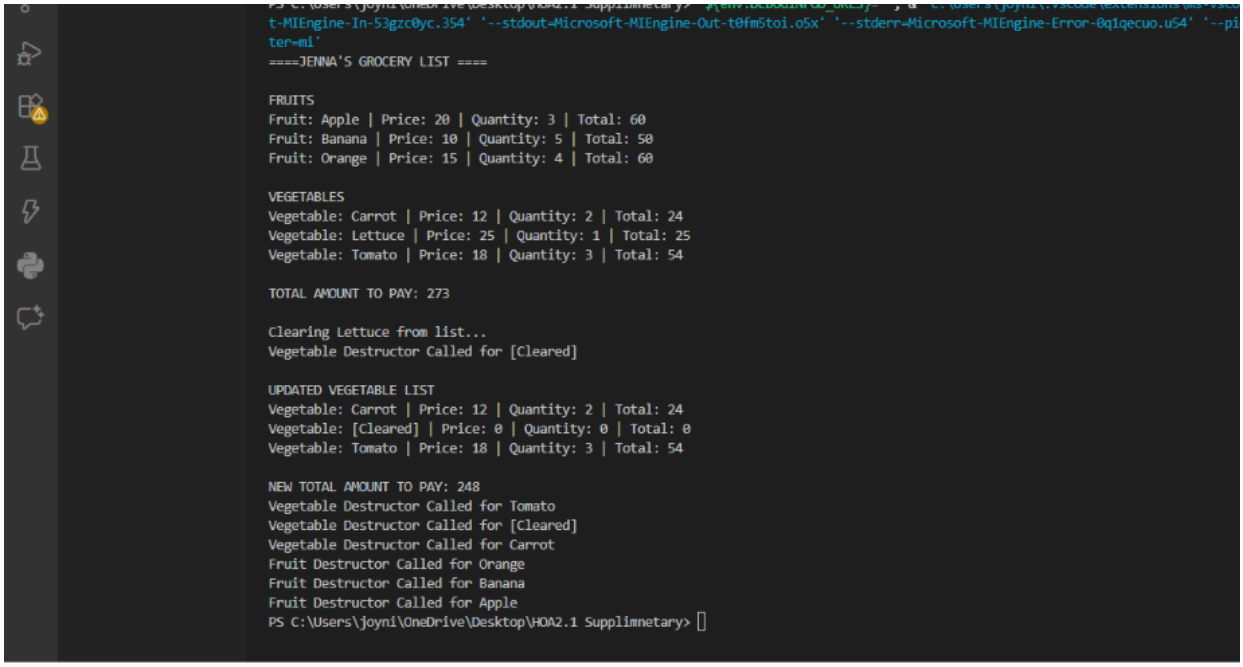
COMPUTER ENGINEERING

QUEZON CITY





OUTPUT



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

```
113     std::cout << "====JENNA'S GROCERY LIST ====" << std::endl;
114
115     std::cout << "\nFRUITS" << std::endl;
116     for (int i = 0; i < 3; i++) {
117         fruits[i].display();
118     }
119
120     std::cout << "\nVEGETABLES" << std::endl;
121     for (int i = 0; i < 3; i++) {
122         vegetables[i].display();
123     }
124
125     std::cout << "\nTOTAL AMOUNT TO PAY: "
126               << TotalSum(fruits, 3, vegetables, 3)
127               << std::endl;
128
129     // Clearing Lettuce
130     std::cout << "\nClearing Lettuce from list..." << std::endl;
131     vegetables[1] = Vegetable("[Cleared]", 0, 0);
132
133     std::cout << "\nUPDATED VEGETABLE LIST" << std::endl;
134     for (int i = 0; i < 3; i++) {
135         vegetables[i].display();
136     }
```

COMPUTER ENGINEERING

QUEZON CITY

This program demonstrates the code in which the Fruit and vegetable classes have attributes name, price, quantity and behavior CalculateSum, display and copy constructor . The program arrays of Fruit and Vegetable object are initialized with sample data.The fruit array contains Apple, Banana, and Orange, and the vegetable array contains Carrot, Lettuce, and Tomato.Each object store each own information, demonstrating how classes can be used to organize related data into a single structure. The program then displays all items of Jenna grocery list.This shows how arrays of objects can be traversed efficiently and how member functions provide a convenient way to access and display object data. Another important observation is the use of the TotalSum()function. This function calculates the overall amount Jenna needs to pay by adding the total cost of every fruit and vegetable item.

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Caplicot and Google**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: To check and to to explain my error and correct the code and for google i used it for translator

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.